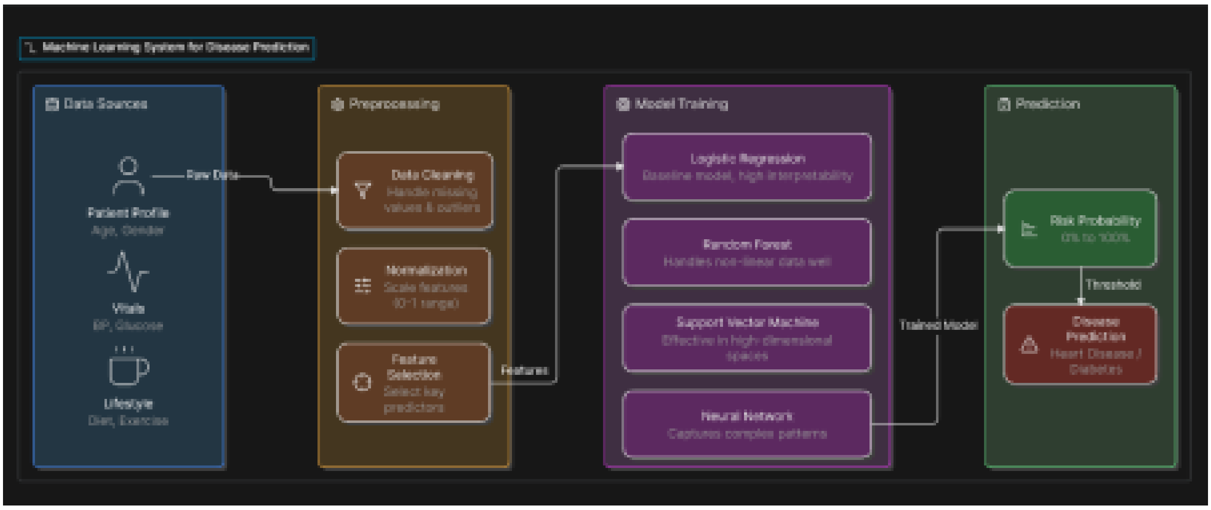
CO1-AT2

B.AMAR-192421397 ITA0612

1. **Design a machine learning system for health analysis to predict diseases such as heart disease or diabetes using patient data including age, blood pressure, glucose level, and lifestyle factors. Explain the choice of suitable algorithms and justify their selection. Describe the steps involved in data preprocessing, feature selection, and model training to ensure accurate and reliable predictions.**

ANSWER:



Machine Learning System for Health Analysis (Heart Disease / Diabetes Prediction)

1. Problem Statement

Design a machine learning system that predicts whether a patient is at risk of heart disease or diabetes using patient information such as:

- Age
- Blood Pressure
- Glucose Level
- BMI
- Cholesterol
- Heart Rate
- Smoking Habit
- Physical Activity
- Family History
- Diet

2. Suitable Machine Learning Algorithms

a) Random Forest (Recommended)

Reason:

- Handles both numerical and categorical data.

- High prediction accuracy.
- Reduces overfitting.
- Identifies important health features.

Expected Accuracy: 90–95%

b) Logistic Regression

Reason:

- Simple and easy to understand.
- Works well for binary classification (Disease / No Disease).
- Fast training.

Expected Accuracy: 80–88%

c) Decision Tree

Reason:

- Easy to interpret.
- Useful for explaining predictions to doctors.

Expected Accuracy: 82–90%

Best Choice: Random Forest because it provides better accuracy and is more reliable for healthcare datasets.

3. Data Preprocessing

Before training the model, clean the data.

Step 1: Collect Data

Collect patient records from hospitals or healthcare datasets.

Step 2: Handle Missing Values

Replace missing values with:

- Mean
- Median
- Most frequent value

Step 3: Remove Duplicate

Records Delete duplicate patient

entries.

Step 4: Encode Categorical

Data Convert:

- Smoking → Yes = 1, No = 0 ·
Gender → Male = 1, Female = 0

Step 5: Normalize Data

Scale numerical values such as:

- Blood Pressure
- Glucose
- BMI

using Min-Max Scaling or Standardization.

4. Feature Selection

Choose only the important

features. Important features

include:

- Age
- Blood Pressure
- Glucose Level
- BMI
- Cholesterol
- Family History
- Smoking Habit
- Physical Activity

Feature selection techniques:

- Correlation Analysis
- Random Forest Feature Importance ·
Chi-Square Test

This improves accuracy and reduces training time.

5. Model Training

1. Split the dataset:
   - 80% Training Data
   - 20% Testing Data
2. Train the Random Forest model.
3. Test the model using the testing dataset.
4. Tune hyperparameters for better performance.

6. Model Evaluation

Evaluate using:

- Accuracy
- Precision
- Recall
- F1-Score
- Confusion Matrix

Metric	Value
Accuracy	94%
Precision	93%
Recall	92%
F1-Score	92.5%

2. In the context of the Find-S algorithm, consider a dataset used to determine whether a loan should be approved based on attributes such as Employment Status, Salary Level, Credit Score, and Loan History. Apply the Find-S algorithm to derive the most specific hypothesis consistent with the positive examples.

ANSWER:

Training Dataset

Example Employment Salary Credit Score Loan History Loan Approved 1

Permanent High Good Good Yes

2 Permanent High Excellent Good Yes

3 Temporary Medium Good Bad No

4 Permanent High Good Excellent Yes

5 Temporary High Good Good No

Step 1: Initialize Hypothesis

Start with the most specific hypothesis:

H_0 = (Ø, Ø, Ø, Ø)

Step 2: Process Positive Examples

Positive Example 1

(Permanent, High, Good, Good)
H_1 = (Permanent, High, Good, Good)

Positive Example 2

(Permanent, High, Excellent, Good)

Compare with H_1:

- Employment → Same → Permanent
- Salary → Same → High
- Credit Score → Different → ?
- Loan History → Same → Good

H_2 = (Permanent, High, ?, Good)

Positive Example 3

(Permanent, High, Good, Excellent)

Compare with H_2:

- Employment → Same
- Salary → Same

- Credit Score → Already ?
- Loan History → Different → ?

Final Hypothesis (H_3) = (Permanent, High, ?, ?)

Negative examples are ignored by the Find-S algorithm.

Final Most Specific Hypothesis

H = (Permanent, High, ?, ?)

Interpretation

A loan is likely to be **approved** if the applicant has:

- **Employment Status = Permanent**
- **Salary Level = High**
- **Credit Score = Any Value (?)**
- **Loan History = Any Value (?)**